AP PRECALCULUS

Unit 1 • Worksheet 1.7-1.11B

Review 1.7-1.11B • Review of Topics 1.7 to 1.11

Name: _____ Date: _____

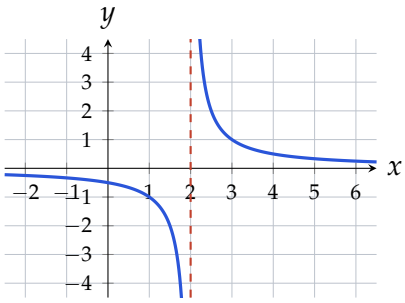
Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1 (1.9 • graph)

NO CALCULATOR

The graph of a rational function s has a vertical asymptote $x = 2$, horizontal asymptote $y = 0$, and passes through $(3, 1)$ and $(1, -1)$. Which could be $s(x)$?



(A) $\frac{1}{x+2}$

(B) $\frac{1}{x-2}$

(C) $\frac{x}{x-2}$

(D) $\frac{x-2}{x}$

Question 2 (1.11)

NO CALCULATOR

$\lim_{x \to -3} \frac{x^2 + x - 6}{x^2 - 9}$ is

(A) $\frac{5}{6}$

(B) 0

(C) $-\frac{5}{6}$

(D) It does not exist.

Question 3 (1.7) NO CALCULATOR

For large positive x, $s(x) = \dfrac{x^2 + x - 1}{x - 1}$ is closest to which line?

(A) $y = x$
(B) $y = x + 1$
(C) $y = x + 2$
(D) $y = 1$

Question 4 (Free response • AP style) NO CALCULATOR

Let $r(x) = \dfrac{ax^2 + bx + 6}{x^2 - 2x - 3}$, where a and b are constants. The graph of r has a horizontal asymptote $y = 2$ and an x-intercept at $(1, 0)$.

(A) Find a and b.
(B) Show that the graph has a hole and find its coordinates.
(C) Find the vertical asymptote and solve $r(x) < 0$.